◇ CAPSTONE II · TAIF UNIVERSITY · 2026 ◇

EQAAB

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Intelligent Autonomous Aerial Surveillance System

ENGINEERING TEAM

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02 BRIEFING OUTLINE

Agenda / 6 chapters

A complete walkthrough of design, build, test, and field validation of the Eqaab platform.

PART ONE OF SIX

01

/ 06

PROBLEM_DOMAIN

Problem Domain

The need, the gap, and the prior work that frames Eqaab.

03 WHERE WE LEFT OFF → WHERE WE ARE

Project Recap

Capstone I delivered a simulated prototype. Capstone II delivers a deployed, real-hardware system flying real missions.

PHASE SHIFT

SIM → FIELD

◇ CAPSTONE I · 2025

Concept & Simulation

- **Gazebo simulation**
  Virtual drone, autonomous waypoint flight in SITL
- **Basic YOLO prototype**
  Off-the-shelf detection on recorded footage
- **System design on paper**
  Block diagrams, sensor selection, requirements
- **No physical hardware**
  No flight, no live communication, no real model

• CAPSTONE II · 2025 / 2026

Built. Trained. Flying.

- **Real assembled drone**
  Carbon-fiber airframe, Pixhawk + RPi5 + Pi Cam
- **Full ROS 2 software stack**
  Mission FSM, detection, IFF, alert publisher
- **Trained YOLOv8 model**
  Custom dataset, 94.3% mAP@50, 8.3 FPS edge
- **Deployed cloud backend**
  FastAPI + PostgreSQL + WSS over 4G LTE
- **Live GCS dashboard**
  React + Leaflet, real-time alerts & commands

04 WHY THIS MATTERS

The Problem

Today's wide-area surveillance is built on fixed sensors and tired human operators — both fail exactly when threats become unconventional.

FAILURE MODES
4 IDENTIFIED

01 Fixed-camera blind spots

Static CCTV cannot cover wide, irregular terrain — borders, holy-site perimeters, remote remote infrastructure. Every fixed installation creates predictable, exploitable gaps in gaps in coverage.

02 Operator fatigue

Human watch-standers monitoring dozens of feeds miss critical events as attention degrades over hours. Vigilance decrements are well-documented and unavoidable.

03 Radar can't see small UAVs

Traditional radar is tuned for large aircraft. Small, low-altitude drones present a radar cross-radar cross-section close to noise — they pass under conventional air defense entirely entirely undetected.

04 Piloted UAVs replicate the fatigue problem

Existing deployed UAV solutions require dedicated human pilots per airframe — recreating recreating the same fatigue and scaling bottleneck the system was meant to eliminate. eliminate.

05 PRIOR ART & RESEARCH GAP

Literature Review

Each related work covers a slice. None integrate detection, autonomy, GPS-tagged alerts, IFF, and a live operator interface into one cohesive system.

STUDIES REVIEWED
5 + 1

REFERENCE	VEHICLES	HUMANS	DRONES	AUTONOMOUS	GPS	GCS UI	IFF
Yun et al. — Multi-UAV RL	—	—	—	✓	✓	—	—
Guettala et al. — Thermal human detection	—	✓	—	—	—	—	—
Yasmine et al. — Anti-drone YOLOv7	—	—	✓	—	—	—	—
Aydin & Singha — Drone detection YOLOv5	—	—	✓	—	—	—	—
Ghazlane et al. — YOLOv7 + DeepSort	✓	✓	—	—	—	—	—
EQAAB (this work)	✓	✓	✓	✓	✓	✓	✓

Research gap: no published work delivers an end-to-end autonomous surveillance UAV with multi-class onboard detection, onboard detection, IFF classification, and an operator-facing GCS — all on commodity hardware.

PART TWO OF SIX

02

/ 06

SOLUTION_OVERVIEW

Solution Overview

What we set out to build, who it serves, and how it fits together.

06 WHAT SUCCESS LOOKS LIKE

Objectives & KPIs

Five measurable goals, each tied to a verifiable performance indicator and an evidence source.

KPIS

5 / 5

01	**Autonomous mission execution**	A complete flight loop — takeoff, waypoint traversal, observation, and observation, and return-to-home — executed end-to-end with **zero zero operator input**.	FSM · WAYPOINTS · RTH
02	**Onboard real-time detection**	Sustained inference at **≥ 5 FPS** on the edge device with no cloud round-trip. round-trip. Detection latency must stay bounded even under thermal load thermal load over a full mission.	RPI5 · YOLOV8N
03	**IFF classification**	Every drone-class detection routed through a deterministic Friendly-vs-Unknown decision with **fail-safe** defaults toward UNKNOWN on any uncertainty.	REGISTRY LOOKUP
04	**GPS-tagged alerts over 4G LTE**	Detection → alert → GCS render in **under 3 seconds** , end-to-end, including network hops. Payload is structured JSON over WSS — no video streaming.	WSS · JSON · TLS
05	**Live GCS dashboard**	A single operator pane with **live map, telemetry, alert feed, and commands** — updating in real time as the drone flies and detects.	REACT · LEAFLET

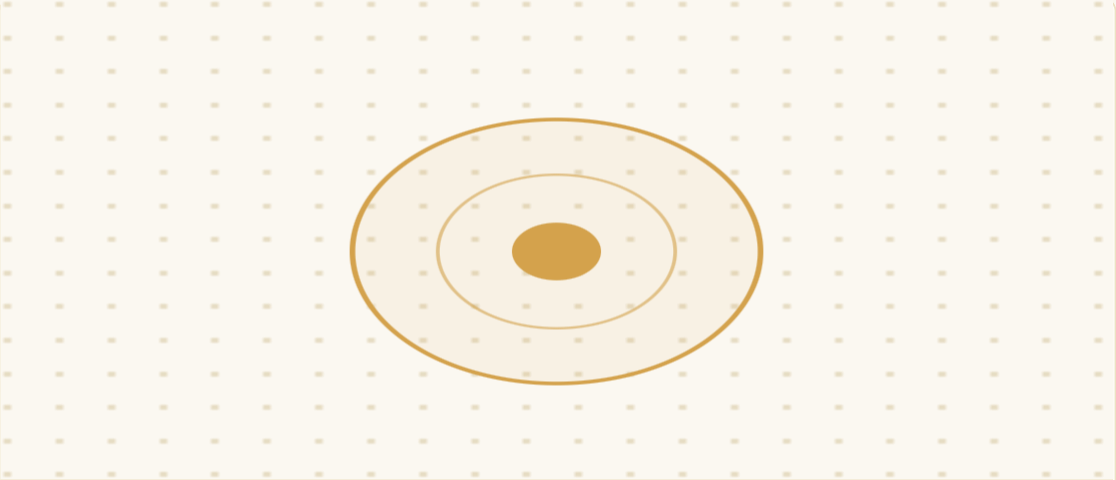
07 WHERE EQAAB EARNS ITS KEEP

Real-World Use Cases

Three deployment scenarios that drove the requirements — and that the platform is built to serve from day one.

MISSION TYPES

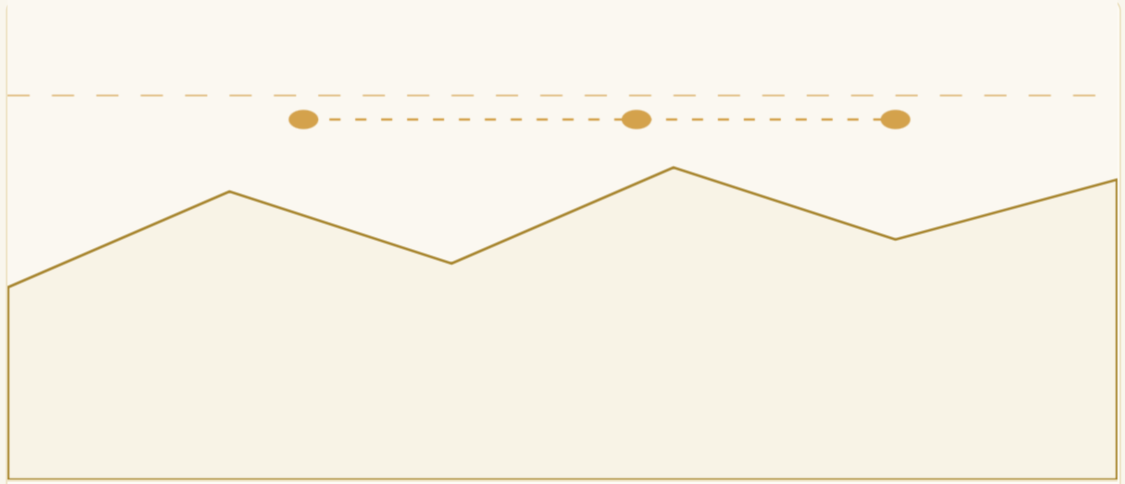
3 PRIMARY



Scenario 01

Hajj Season Security

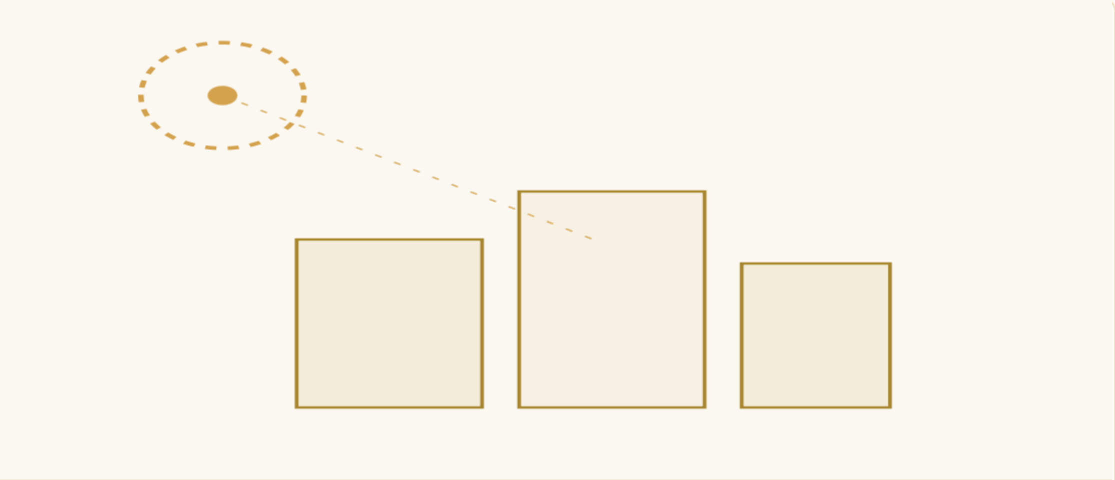
Aerial overwatch of dense pilgrim crowds and unauthorized drone activity around Makkah and the holy the holy sites — coverage no fixed sensor array can can match.



Scenario 02

Border Surveillance

Wide-area perimeter patrol across remote terrain where fixed cameras and radar coverage are economically and operationally infeasible.



Scenario 03

Critical Infrastructure

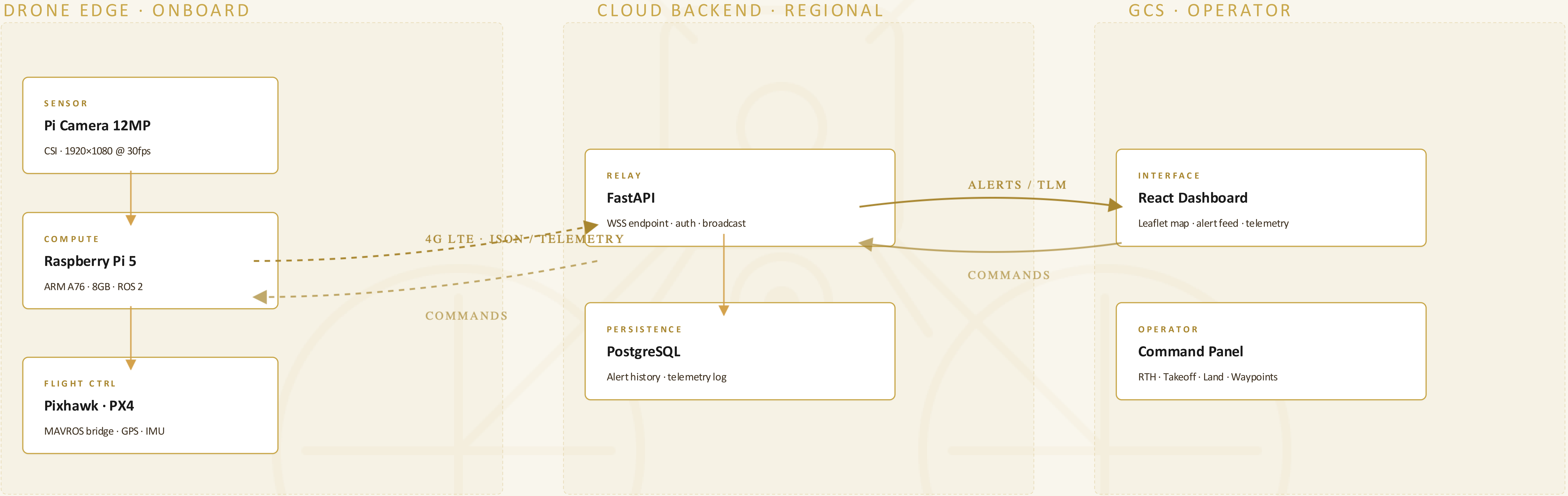
Persistent surveillance over power plants, airports, refineries, and restricted facilities — including detection of small unauthorized UAVs entering protected airspace.

08 END-TO-END SIGNAL FLOW

System **Architecture**

Three tiers — drone edge, cloud backend, operator interface — connected over 4G LTE with structured JSON only.

TIERS
3 LAYERS



DATA POLICY Only structured JSON alerts and telemetry cross the LTE link — no video streaming. Bandwidth requirement drops by approximately **95%**.

PART THREE OF SIX SIX

03

/ 06

DESIGN_AND_BUILD

System Design & Build

From the carbon-fiber airframe to the flight software and ground link.

- 09 Hardware Stack
- 10 ROS 2 Software Stack
- 11 Backend & Comms Comms
- 12 GCS Dashboard

09 BILL OF MATERIALS & FLIGHT ENVELOPE

Hardware Stack

Off-the-shelf components, integrated and tuned. Production-grade behaviour from commodity hardware.

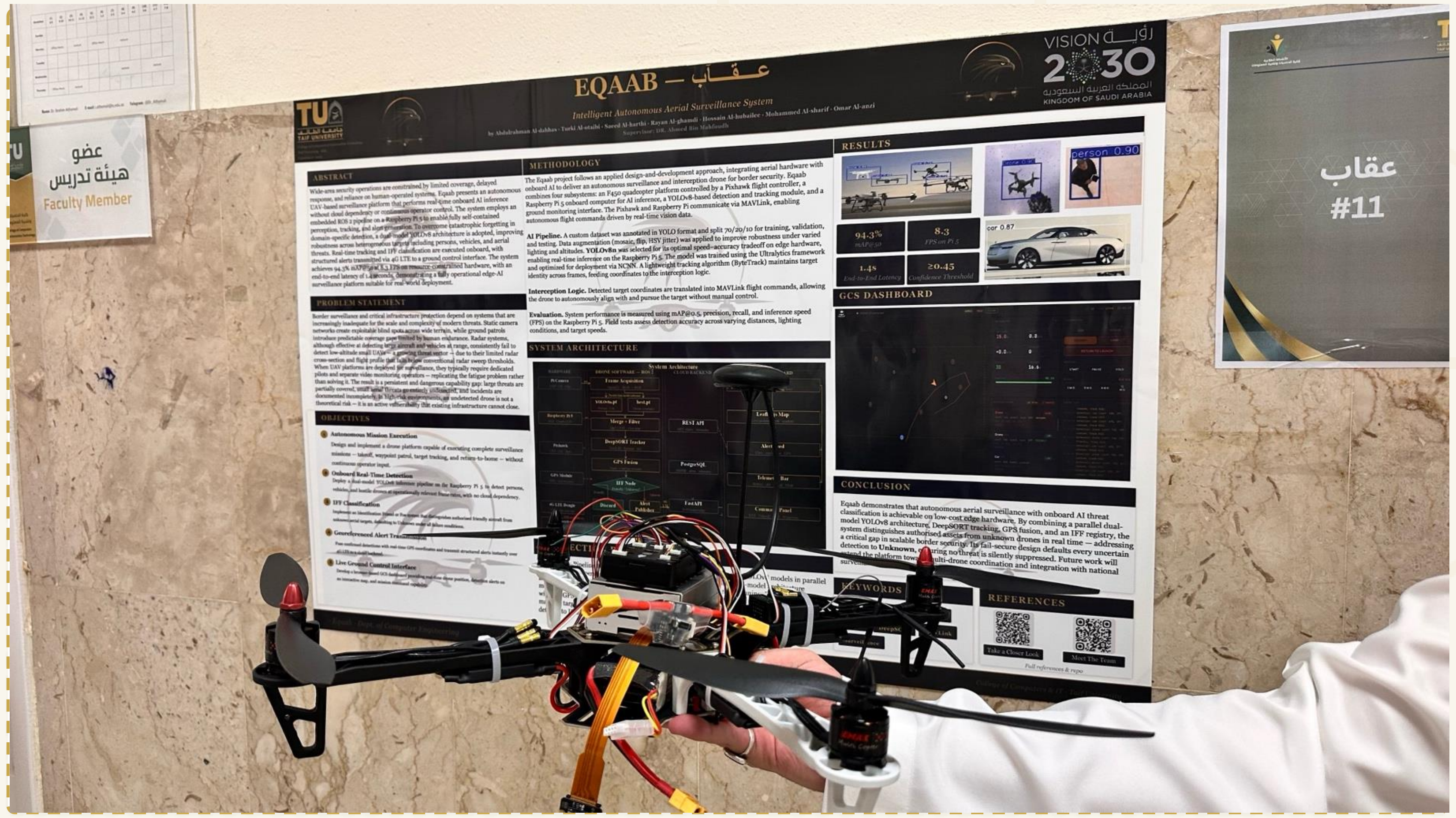
COMPONENTS
8 SUBSYSTEMS

A · BILL OF MATERIALS

Frame	F450 Nylon Composite Quadcopter Frame
Flight Controller	Pixhawk · PX4 firmware · 32-bit ARM
Companion Computer	Raspberry Pi 5 · Quad-core A76 · 8GB RAM
Camera	Pi Camera Module · 12MP CSI
GPS	u-blox M8N · ±2m accuracy
Comms	4G LTE USB dongle · global reach
Battery	zee 3s 5200 mAh
RC Backup	2.4GHz receiver · manual override

B · FLIGHT SPECIFICATIONS

ALTITUDE	FLIGHT TIME	RANGE	GEOFENCE	FAILSAFE
50 m	15–20 min	∞ 4G	Polygon	RTH @ 30%



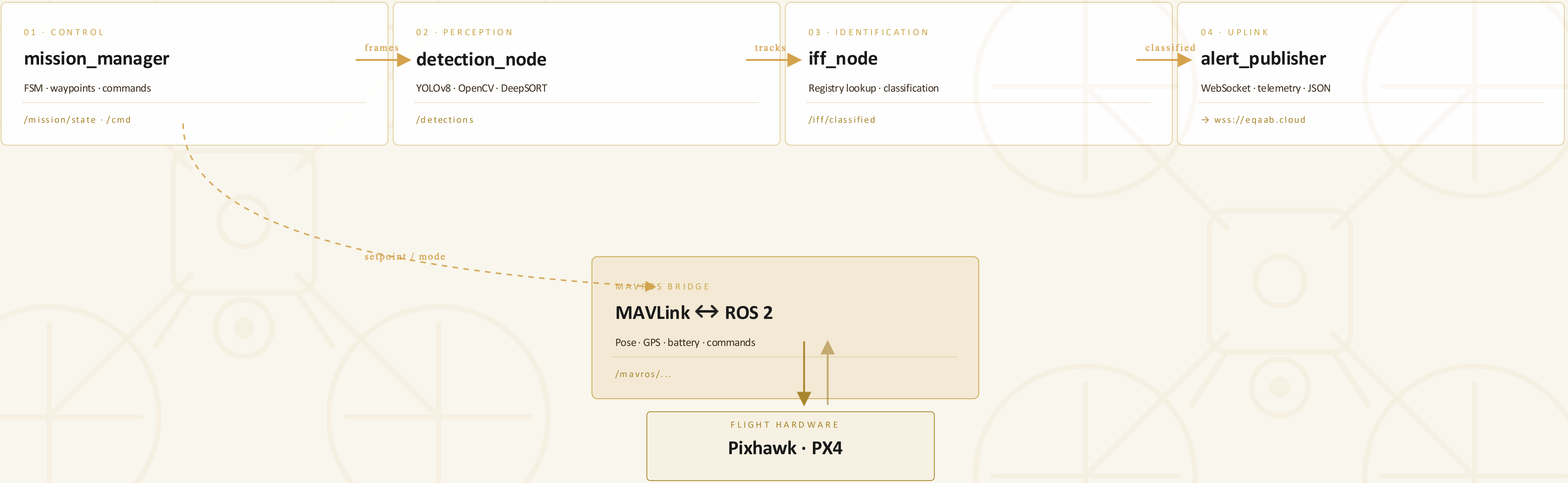
10 SOFTWARE ARCHITECTURE · ONBOARD

ROS 2 Node Graph

NODES

4 + MAVROS

Four cooperating nodes orchestrate mission, perception, identification, and uplink.



11 CLOUD RELAY & PERSISTENCE

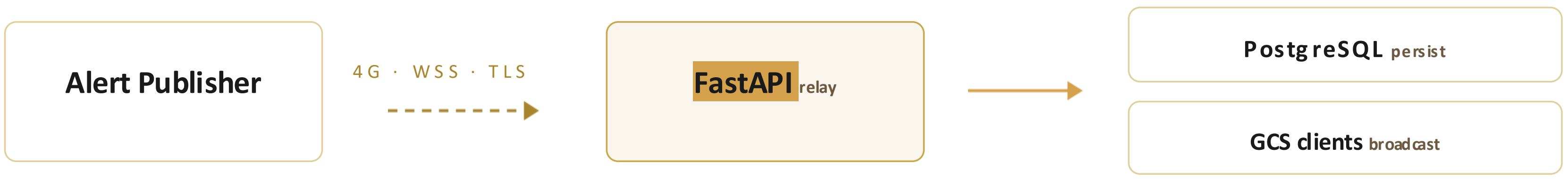
Backend & Communication

A thin, opinionated cloud tier — relay, persist, broadcast. Nothing more.

PROTOCOL

WSS · TLS

ARCHITECTURE



95%

BANDWIDTH SAVED

Structured JSON only — no video streaming over LTE. Cuts uplink requirement to a fraction of comparable telepresence comparable telepresence systems.

MESSAGE · TELEMETRY


```
{
"type": "telemetry",
"drone_id": "EQB-01",
"ts": 1717938472,
"lat": 21.4373,
"lon": 40.5127,
"alt": 47.2,
"battery": 68,
"mode": "AUTO" }
```


MESSAGE · ALERT

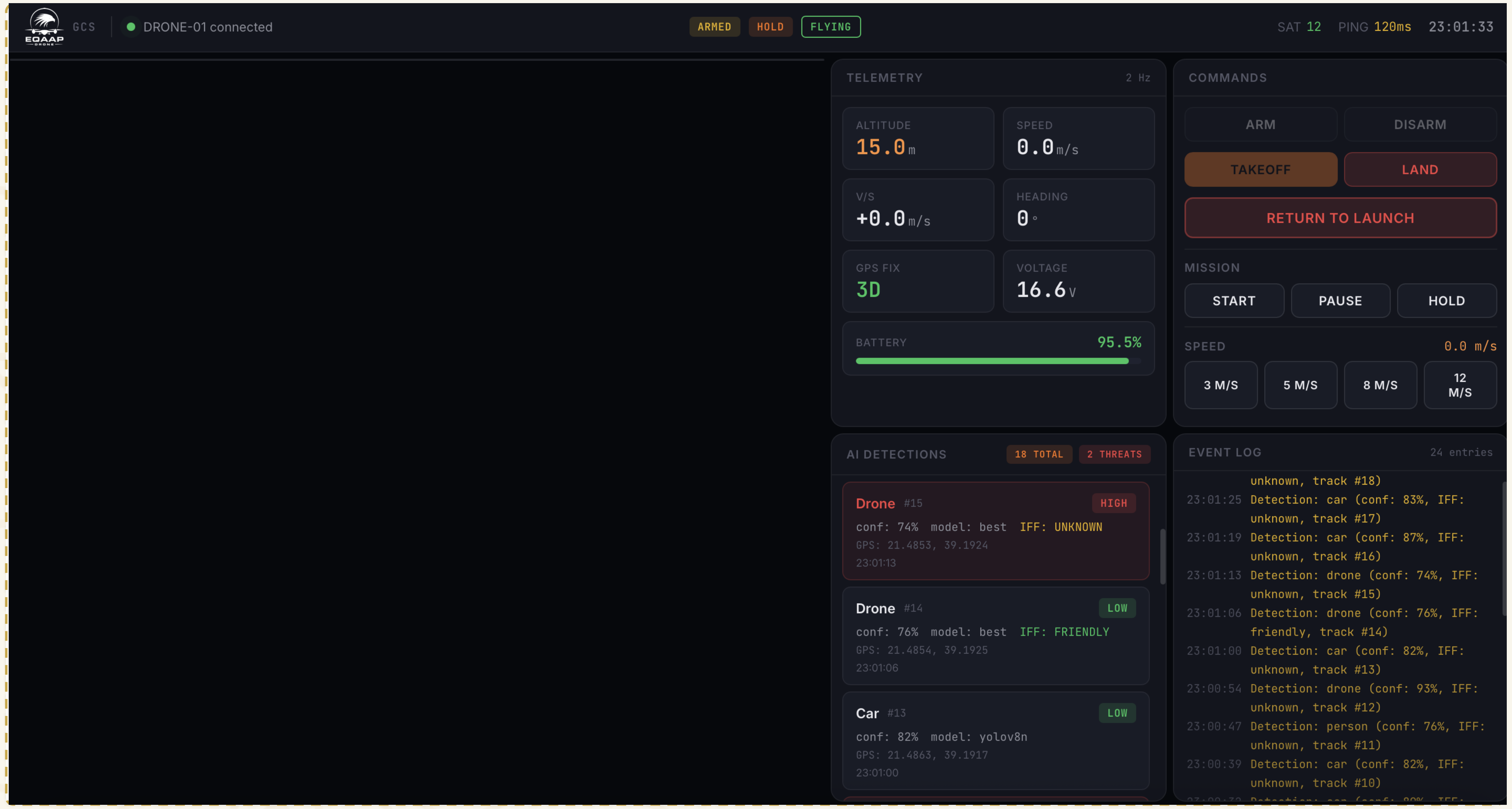

```
{
"type": "alert",
"class": "drone",
"iff": "UNKNOWN",
"lat": 21.4391,
"lon": 40.5103,
"conf": 0.87,
"track_id": 42 }
```

12 THE OPERATOR'S WINDOW

GCS Dashboard

One pane of glass — map, alerts, telemetry, commands. Built with React + Leaflet.

LATENCY
~1.4S E2E



heading

01 **Live Leaflet map**

1Drone position,, and flight path rendered in real time over satellite tiles.

satellite tiles.

02 **Alert feed**

Streaming list of GPS-tagged detections with class, IFF state, and confidence.

03 **Telemetry bar**

Battery · altitude · speed · flight mode at a glance.

04 **Command panel**

RTH · Takeoff · Land · Waypoint upload from the operator's seat.

seat.

PART FOUR OF SIX

04

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INTELLIGENCE_LAYER

Intelligence Layer

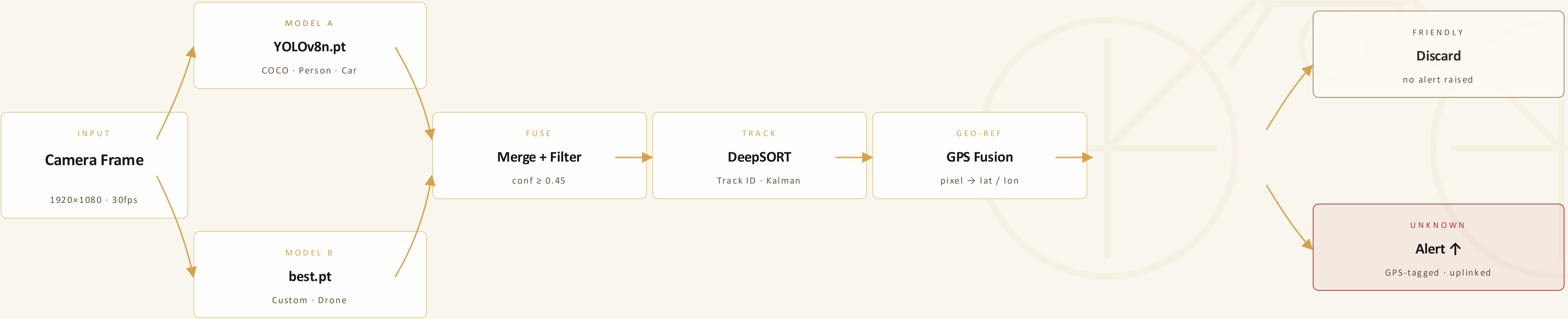
Perception, identification, and the data that taught the system to see.

13 FROM PIXELS TO GPS-TAGGED ALERTS

AI Detection Pipeline

A dual-model perception stack with tracking, GPS fusion, and identity classification — all on the edge device.

MODELS
2 · PARALLEL



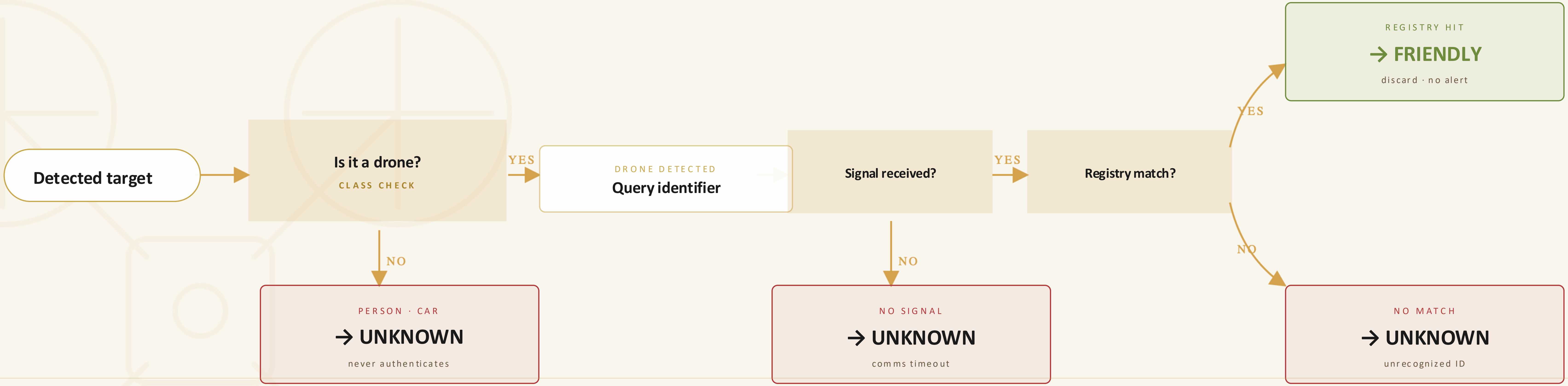
DESIGN NOTE Dual-model architecture adopted to prevent **catastrophic forgetting** — single-model fine-tuning caused measurable degradation of COCO classes during early experiments.

14 IDENTIFICATION FRIEND OR FOE

IFF Decision Logic

A deterministic, fail-safe decision tree. Persons and cars never authenticate. Drones do — or get flagged.

FAIL-SAFE
→ UNKNOWN



FAIL-SAFE

Every failure path — class mismatch, missing signal, registry miss — defaults to **UNKNOWN**. The system raises false alerts rather than miss a threat.
than miss a threat.

15 FROM DATASET TO DEPLOYED WEIGHTS

Training & Dataset

Reproducible pipeline — open dataset, public framework, single GPU. Result is the best.pt shipped to the drone.
drone.

FINAL MAP@50

94.3%

A · DATASET

Composition

Source	Roboflow
Total images	7,500
Classes	3 (drone · person · car)
Split	60 / 20 / 20
Augmentation	flip · brightness · mosaic · crop

B · TRAINING

Configuration

Platform	Kaggle T4 GPU
Framework	Ultralytics YOLOv8
Base model	yolov8n.pt
Epochs	100
Batch	16
Optimizer	AdamW

PART FIVE OF SIX

05

/ 06

VALIDATION_PERF

Validation & Performance

How we tested it. What it actually does in the air.

16 HOW WE PROVED IT WORKS

Testing Methodology

Four stages — from individual ROS nodes up to live field flights — each with its own tooling and pass criteria.

STAGES

4 LEVELS

STAGE	WHAT WAS TESTED	METHOD	TOOL
UNIT	Individual ROS 2 nodes	Isolated node testing	ros2 launch · pytest
INTEGRATION	Full pipeline data flow	End-to-end message tracing	rqt_graph · terminal logs
SIMULATION	Autonomous mission FSM	Virtual flight	Gazebo · ArduPilot SITL
FIELD	Detection accuracy	Real flights · live targets	Physical drone + GCS

Unit

Each ROS node validated in isolation against mocked inputs.

Integration

Topic graph traced from camera frame to alert publish.

Simulation

FSM executed virtual missions in Gazebo before real flight.

Field

Eqaab flown against drone, vehicle, and human targets.

17 WHAT THE PLATFORM ACTUALLY DELIVERS

Results & Performance

Validated across model accuracy, edge inference, and end-to-end alert latency.

STATUS
ALL TARGETS MET

94.3%
MAP @ 0.5
Validation on held-out test split (550 images)

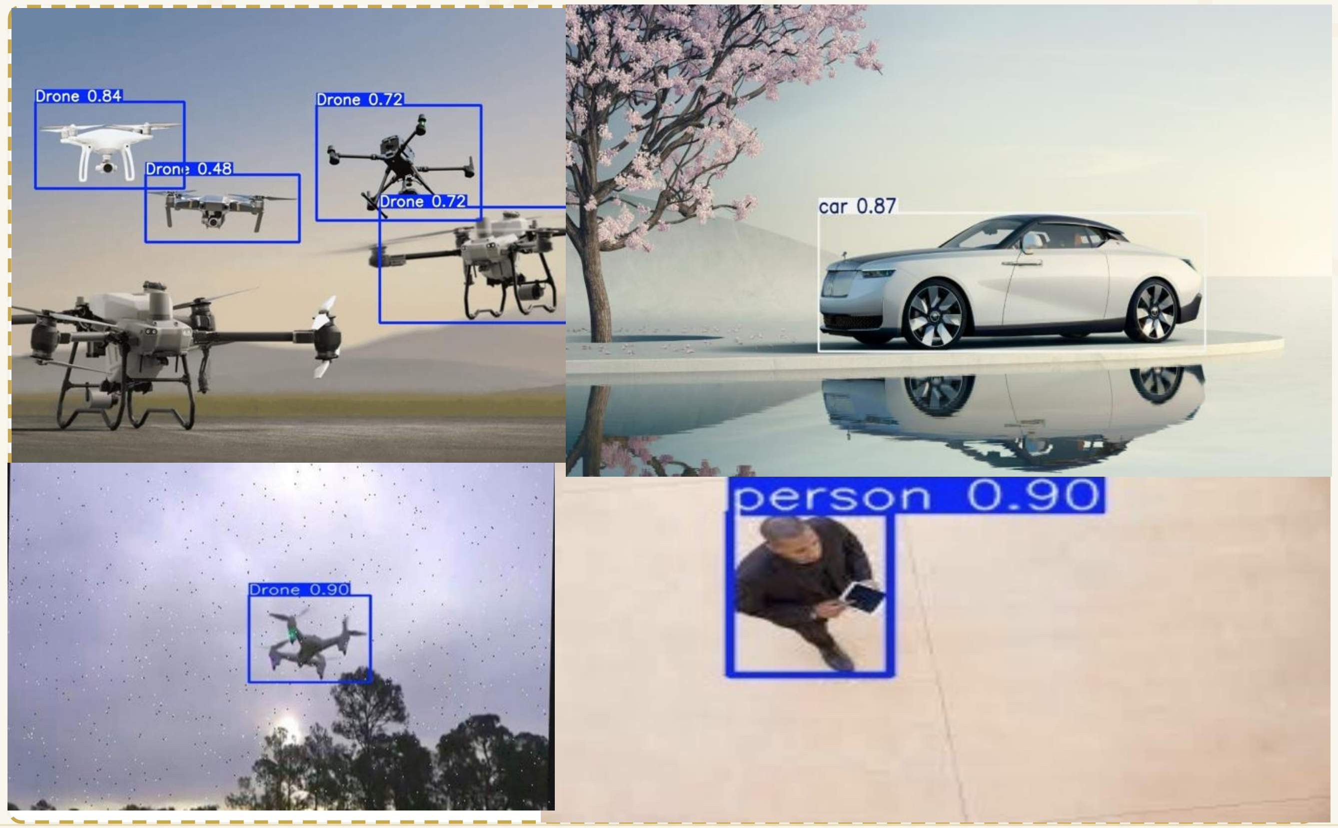
8.3fps
EDGE INFERENCE
On Raspberry Pi 5 · CPU-only · 640px input

91.7%
PRECISION
Averaged across 3 classes, conf ≥ 0.45

1.4s
ALERT LATENCY
Detection → GCS render, measured end-to-end

PER-CLASS BREAKDOWN

CLASS	PRECISION	RECALL	F1	MAP@50
Drone	0.923	0.901	0.912	0.947
Person	0.914	0.889	0.901	0.938
Car	0.915	0.908	0.911	0.944
Macro avg	**0.917**	**0.899**	**0.908**	**0.943**

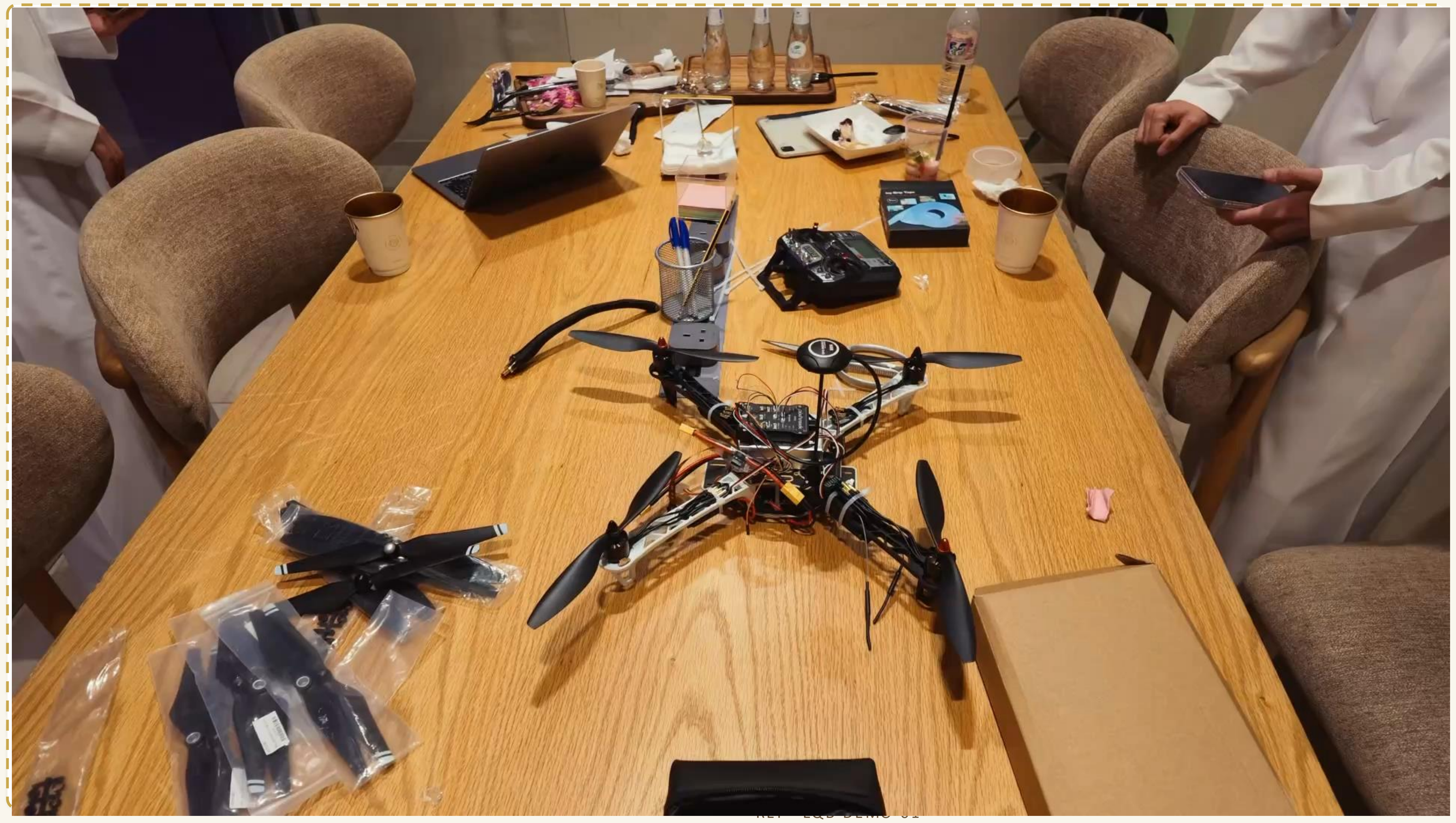


18 THE SYSTEM IN MOTION

Live Demo

Full loop: drone detects target → onboard inference → IFF check → GPS-tagged alert → GCS map updates.

PLAYBACK



A Detection

YOLOv8 identifies a drone target in the live camera frame; DeepSORT DeepSORT assigns a track ID.

B IFF + alert

Identifier query fails → target marked UNKNOWN → JSON alert published over WSS.

C GCS update

Backend relays the alert; React dashboard renders the GPS-tagged tagged marker on the live map.

PART SIX OF SIX

06

/ 06

CONCLUSIONS_FUTURE

Conclusions & Future

Where we landed, what we learned, and where Eqaab goes next.

19 PRODUCTION-GRADE FOR 1% OF LEGACY COST

Cost Analysis

A complete intelligent surveillance platform — airframe through cloud — for less than a single high-end CCTV install.

TOTAL BUILD

2,690 SAR

EQAAB · THIS WORK

2,690 SAR · TOTAL

F450 Nylon Composite Quadcopter Frame	820
Pixhawk + PX4 stack	450
Raspberry Pi 5 (8GB)	380
Pi Camera Module	190
GPS · 4G dongle · battery	540
RC receiver · misc.	310

TRADITIONAL SURVEILLANCE

1,000,000 rs rs SAR

Fixed CCTV + tower install	100k+
Long-range radar	300k+
Piloted UAV system	500k+
Operator staffing (annual)	200k+
Maintenance & comms	recurring

1% Production-grade intelligent surveillance at roughly **1% of traditional infrastructure cost** — without sacrificing autonomy, identification, or operator control. operator control.

20 WHAT WE PROMISED VS WHAT WE DELIVERED

Objectives vs Achieved

Five for five — every committed KPI met or exceeded, each backed by verifiable evidence.

RESULT

5 / 5 ✓

OBJECTIVE	TARGET	ACHIEVED	EVIDENCE
Autonomous flight	Full mission	✓ Verified	Gazebo + field flights
Real-time detection	≥ 5 FPS	✓ 8.3 FPS	RPi5 benchmark · 640px
IFF classification	Friendly / Unknown	✓ Operational	Registry lookup tested
Alert transmission	< 3s latency	✓ ~1.4s	End-to-end measurement
GCS dashboard	Live map + commands	✓ Deployed	React dashboard live

5
OBJECTIVES SET

5
OBJECTIVES MET

2
EXCEEDED TARGET

Detection FPS exceeded target by **66%**; alert latency came in at less than **half** the requirement. requirement.

21 WHAT BROKE — AND HOW WE FIXED IT

Challenges & Solutions

Three real engineering problems that shaped the final architecture.

PIVOTS
3 MAJOR

01

CHALLENGE

Catastrophic forgetting

Fine-tuning a single YOLOv8 model on the custom drone class measurably degraded its accuracy on accuracy on the COCO person and car classes — making the model worse at the things it was already was already good at.

→

SOLUTION

Dual-model parallel architecture

Run the stock yolov8n.pt and the custom best.pt in parallel on every frame, then merge merge and filter. Each model stays specialised; nothing forgets.

02

CHALLENGE

4G LTE latency for piloting

Manual remote piloting over cellular incurs latency too high for safe low-altitude flight — control-loop lag became unacceptable as soon as we left line of sight.

→

SOLUTION

Fully autonomous FSM

Eliminate the manual control loop entirely. Operator issues *high-level* commands (RTH, takeoff, waypoints); the FSM executes locally with onboard sensors at deterministic rates.

03

CHALLENGE

CPU-only inference on RPi5

No GPU, no NPU. Larger YOLOv8 variants stalled below 3 FPS — well under our 5 FPS KPI — and risked thermal throttling during long missions.

→

SOLUTION

Nano variant + frame-rate cap

Switched to YOLOv8 nano, capped the inference loop, and kept the input at 640px. Sustained **8.3 FPS** with stable thermals and headroom for tracker overhead.

22 WHERE EQAAB GOES NEXT

Future Work

Four directions that build directly on the v1 platform — none require throwing the architecture away.

ROADMAP

V2 · V3

V2 · COORDINATION

Multi-drone Coordination

Multiple Eqaab airframes sharing a single GCS — unified alert feed, deconflicted patrol

patrol routes, and cooperative tracking of fast-moving targets.

V2 · SENSOR

Thermal Imaging

Add a thermal payload alongside the RGB camera. Detection at night, through smoke, and against camouflaged targets — same pipeline, fused channels.

V3 · SECURITY

Cryptographic IFF

Replace proximity-based identifier matching with hardware-authenticated transponders.

transponders. Public-key challenge / response — spoof-resistant by design.

V2 · PERFORMANCE

Edge Optimization

INT8 quantization and structured pruning of the YOLOv8 weights. Target: double the edge

the edge FPS without dropping mAP — opens headroom for larger backbones.

23 WHAT WE BUILT · AND WHO BUILT IT

Conclusion

STATUS

DELIVERED

“ Eqaab demonstrates that a seven-member student engineering team can engineering team can design, build, and validate a complete production-production-grade intelligent aerial security platform — integrating integrating autonomous flight, onboard AI, IFF classification, and live live operator interfaces into a single cohesive system at a fraction of of traditional surveillance cost. ”

7	24	1	94.3%
STUDENTS	WEEKS	FLYING DRONE	mAP @ 50

THE TEAM

Abdulrahman Al-dahhas	**Team Lead** · AI Engineer &
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Saeed Al-harthi	Network Engineer & CyberSecurity CyberSecurity
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Meet Eqaab's Team

◇ THANK YOU ◇

Take Clooser look